

JEPA-FORGE

Expanded prototype: 23 datasets

138 completed GPU runs

120 new runs across 20 additional datasets

The original three-dataset, 18-run study is preserved as a historical snapshot. The expansion adds 10 public datasets and 10 synthetic mechanisms, with two task policies and three fixed model seeds for every dataset.

This report measures a small JEPA-style task-compilation and representation-learning pilot. It does not claim official benchmark reproduction or general superiority.

Observed primary-task comparisons

JEPA compared with	Wins	Ties	Losses
Raw linear	9	0	11
Extra Trees	3	0	17
Random encoder	9	0	11

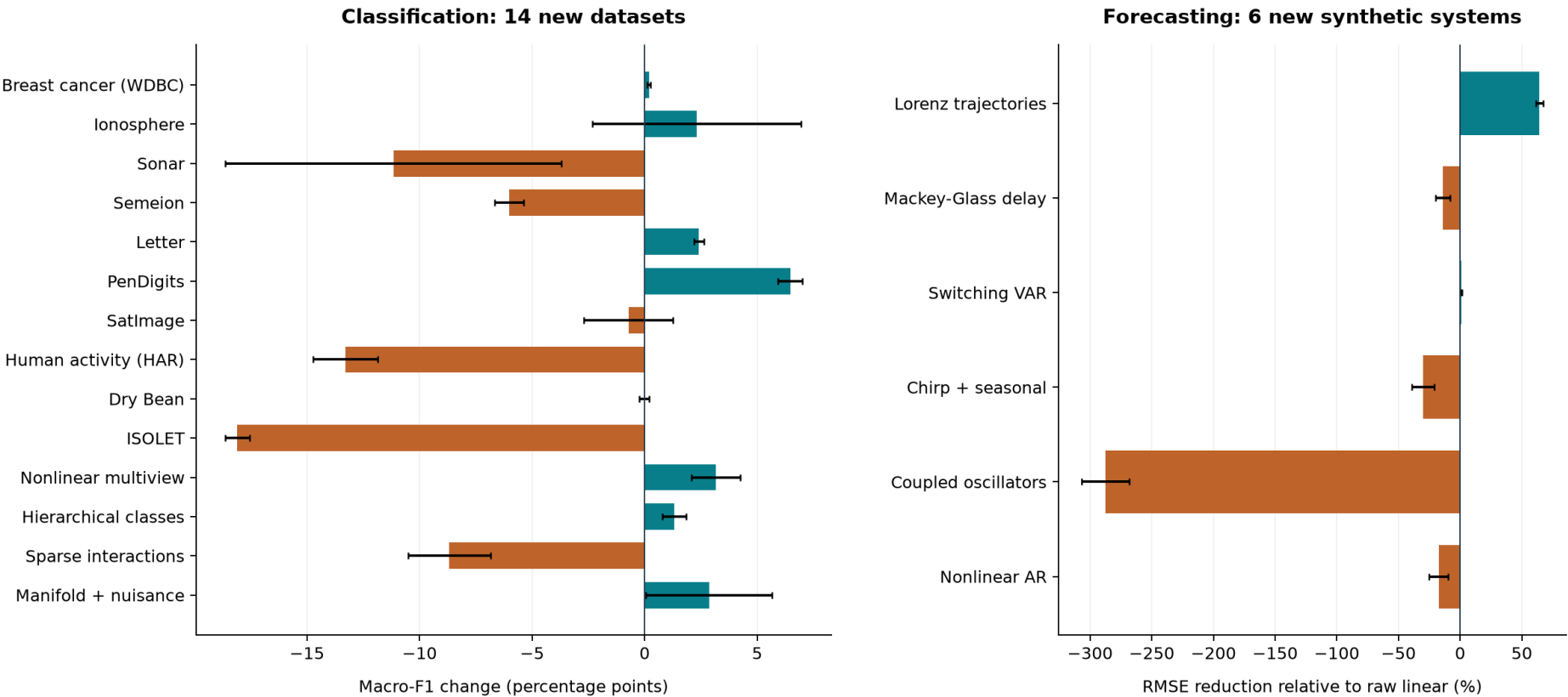
Counts compare the seed-mean test metric within each of 20 predeclared primary tasks. Classification uses higher macro F1; forecasting uses lower RMSE. Counts are descriptive and are not significance tests.

Low-rank/low-variance heuristic: 83 of 120 new runs flagged; 90 of 138 combined runs. A flag requires investigation and is not proof that every representation is constant.

Evidence: 40 export reloads, 120 checkpoint hashes and CPU replays; 120 tests passed. Neural device evidence is mps:0. Classical estimators and validation replay use CPU.

Results across all 20 new primary tasks

JEPA relative to the matched raw-context linear baseline



Positive values favor JEPA. Primary policies fixed before evaluation. Bars: three-seed means; whiskers: one model-seed SD, not confidence intervals.
RMSE is compared within each system; percentages do not average across systems. Extra Trees and other baselines appear in the report tables.

Positive values favor JEPA relative to the matched raw-context linear baseline. Classification changes are macro-F1 percentage points; forecast reductions are percentages of each system's raw-linear RMSE. Whiskers show one model-seed SD, not confidence intervals. Cross-generator RMSE values and percentage effects are not pooled. Extra Trees and other comparators remain in the report tables.

Protocol and verification

Seeds 7, 17, 27; split/generator seed 2026; 60 fixed epochs; the final epoch is retained. First declared policy is primary before any test evaluation.

Training-only feature scaling and PCA; separate labels for probes. Validation macro F1 or RMSE selects probe hyperparameters; the selected estimator remains fitted on training rows. Test observations do not select models, policies, or hyperparameters.

Context-only comparisons: raw linear model, PCA plus linear probe, Extra Trees, random encoder plus linear probe, and trained JEPA encoder plus linear probe. Full-input classification references are separately labeled; forecasting also uses persistence.

Logistic C and ridge alpha use 0.01, 0.1, 1, 10, 100. Extra Trees uses 100 trees and depth 8 or unlimited. PCA uses at most eight components. The JEPA pilot uses latent dimension 32, batch size 128, Adam 0.001, EMA 0.99, and variance weight 1.0.

Reported dispersion is sample standard deviation across three model seeds on one fixed dataset split. Deterministic methods can have zero seed variation. These are optimization repeats, not independent data replications.

GPU evidence includes parameters, batches, loss, and online gradients on mps:0, with a frozen EMA teacher and no silent CPU fallback. Export reloads validate schemas, splits and checksums. Independent checkpoint replay uses the first 32 training rows on CPU with absolute and relative tolerance 1e-4.

Current source/configuration hashes are validated for the expansion. The original runtime is archived separately; its old measurements are not misrepresented as reruns under today's code.

Scope limits

Public datasets are bounded samples with new internal partitions; they are not original competition or UCI benchmark splits. Known entities are held out where identifiers are available. Synthetic equations are controlled stress tests, not calibrated scientific simulators. No clinical-utility, physical-validity, deployment, human-study, or broad state-of-the-art claim follows from this pilot.

Public dataset catalog

Dataset / mechanism	Rows x features	Groups	Structural challenge
Breast cancer (WDBC)	569 x 30	569	Thirty correlated morphology measurements across mean, standard-error and worst-value summaries; small real biomedical classification set.
Ionosphere	350 x 34	rows	Seventeen complex-valued radar pulse pairs, mixed constant and informative channels, and nonlinear signal structure.
Sonar	208 x 60	rows	Sixty ordered spectral energy bands with only 208 observations; a high-feature-to-sample real sensing problem.
Semeion	1,593 x 256	rows	256 binary image pixels and ten classes at 16x16 resolution, larger spatial masks than the original 8x8 Digits pilot.
Letter	3,000 x 16	rows	Twenty-six classes with geometric moments and edge summaries extracted from distorted font images.
PenDigits	2,992 x 16	44	Eight ordered two-coordinate trajectory points; ten classes and preserved source grouping across 44 writer-like groups.
SatImage	3,000 x 36	rows	Nine spatial neighbors times four spectral bands; six land-cover classes and interdependent spatial/spectral features.
Human activity (HAR)	3,000 x 561	30	561 engineered inertial time/frequency features with six activities and explicit subject-held-out evaluation.
Dry Bean	3,000 x 16	rows	Seven varieties with correlated geometric and shape features, imbalance and redundant derived attributes.
ISOLET	3,000 x 617	rows	617 acoustic summary features with 26 spoken-letter classes; substantially higher dimensional than the initial benchmark.

Exact source URLs, attribution, generator parameters, split sizes, raw-array hashes, and all policy definitions appear in the Markdown report and exported manifests. Public dataset metadata record CC BY 4.0; synthetic outputs record CC0-1.0. Software licensing is separate.

Synthetic mechanism catalog

Dataset / mechanism	Rows x features	Groups	Structural challenge
Lorenz trajectories	2,000 x 96	100	Lorenz differential equations integrated by RK4; independent initial states/rho and noisy three-state observations
Mackey-Glass delay	2,000 x 96	100	Noisy discrete Euler Mackey-Glass-inspired delayed feedback with three causal observation channels
Switching VAR	2,000 x 128	100	Three-regime four-channel vector autoregression with persistent latent Markov switching and entity-specific noise scale
Chirp + seasonal	2,000 x 96	100	Nonstationary chirp phase plus slow amplitude modulation, fixed seasonal component, linear trend and colored noise
Coupled oscillators	2,000 x 128	100	Two bidirectionally coupled damped cubic oscillators with independent driving phases and noisy semi-implicit Euler integration
Nonlinear AR	2,000 x 96	100	Three-channel nonlinear autoregression with multiplicative cross-lag interactions and state-dependent innovation variance
Nonlinear multiview	2,000 x 64	rows	Two nonlinear noisy measurements of six shared Gaussian latent factors; three evaluation classes from latent nonlinear scores
Hierarchical classes	2,000 x 80	rows	Six latent Gaussian subclasses nested under three separated parent centers, observed through linear and nonlinear views
Sparse interactions	2,000 x 96	rows	Sparse nonlinear classification signal among 70 independent nuisance columns; source factors and pairwise products occupy separate views
Manifold + nuisance	2,000 x 96	rows	Rolled two-dimensional manifold observed through two different embeddings with 64 independent-of-label correlated nuisance measurements

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New primary classification tasks

Mean +/- sample SD, three seeds. Macro F1 is a fraction; higher is better.

Dataset	JEPA	Raw linear	Extra Trees	Random encoder
Breast cancer (WDBC)	0.921 +/- 0.001	0.919 +/- 0.000	0.899 +/- 0.012	0.908 +/- 0.006
Ionosphere	0.770 +/- 0.046	0.746 +/- 0.000	0.896 +/- 0.015	0.811 +/- 0.057
Sonar	0.597 +/- 0.075	0.708 +/- 0.000	0.809 +/- 0.024	0.704 +/- 0.053
Semeion	0.849 +/- 0.006	0.909 +/- 0.000	0.908 +/- 0.007	0.722 +/- 0.021
Letter	0.522 +/- 0.002	0.498 +/- 0.000	0.700 +/- 0.011	0.602 +/- 0.012
PenDigits	0.898 +/- 0.005	0.833 +/- 0.000	0.911 +/- 0.001	0.901 +/- 0.001
SatImage	0.799 +/- 0.020	0.806 +/- 0.000	0.868 +/- 0.008	0.839 +/- 0.010
Human activity (HAR)	0.796 +/- 0.014	0.929 +/- 0.000	0.916 +/- 0.001	0.789 +/- 0.008
Dry Bean	0.914 +/- 0.002	0.914 +/- 0.000	0.916 +/- 0.003	0.923 +/- 0.008
ISOLET	0.748 +/- 0.005	0.929 +/- 0.000	0.932 +/- 0.006	0.703 +/- 0.013
Nonlinear multiview	0.880 +/- 0.011	0.849 +/- 0.000	0.919 +/- 0.007	0.888 +/- 0.018
Hierarchical classes	0.963 +/- 0.005	0.950 +/- 0.000	0.961 +/- 0.003	0.952 +/- 0.003
Sparse interactions	0.578 +/- 0.018	0.665 +/- 0.000	0.657 +/- 0.011	0.600 +/- 0.027
Manifold + nuisance	0.765 +/- 0.028	0.736 +/- 0.000	0.958 +/- 0.003	0.859 +/- 0.004

The primary policy was fixed in advance. Both policies and all six methods are retained in the complete Markdown appendix. Full-input classification references have more information and are not matched baselines; forecast persistence uses only the last observation.

New primary forecasting tasks

Mean +/- sample SD, three seeds. RMSE is in original generator units; lower is better. Do not compare RMSE across different generators.

Dataset	JEPA	Raw linear	Extra Trees	Random encoder
Lorenz trajectories	0.850 +/- 0.072	2.384 +/- 0.000	0.589 +/- 0.002	1.885 +/- 0.094
Mackey-Glass delay	0.089 +/- 0.005	0.078 +/- 0.000	0.087 +/- 0.001	0.094 +/- 0.007
Switching VAR	0.183 +/- 0.001	0.185 +/- 0.000	0.179 +/- 0.000	0.188 +/- 0.000
Chirp + seasonal	0.211 +/- 0.015	0.162 +/- 0.000	0.197 +/- 0.000	0.257 +/- 0.010
Coupled oscillators	0.138 +/- 0.007	0.036 +/- 0.000	0.131 +/- 0.001	0.079 +/- 0.004
Nonlinear AR	0.115 +/- 0.008	0.098 +/- 0.000	0.121 +/- 0.000	0.107 +/- 0.002

The primary policy was fixed in advance. Both policies and all six methods are retained in the complete Markdown appendix. Full-input classification references have more information and are not matched baselines; forecast persistence uses only the last observation.

All 40 new policies: key comparisons

Mean +/- sample SD across three seeds. F1 denotes macro F1 (higher is better); RMSE is lower-is-better. The Markdown report retains PCA, random-encoder, and full-input/persistence rows as well.

Dataset	Policy (* primary)	Metric	JEPA	Raw linear	Extra Trees
Breast cancer (WDBC)	mean_to_worst *	F1	0.921 +/- 0.001	0.919 +/- 0.000	0.899 +/- 0.012
Breast cancer (WDBC)	mean_to_error	F1	0.921 +/- 0.001	0.919 +/- 0.000	0.899 +/- 0.012
Ionosphere	alternating_pairs *	F1	0.770 +/- 0.046	0.746 +/- 0.000	0.896 +/- 0.015
Ionosphere	first_half	F1	0.798 +/- 0.018	0.833 +/- 0.000	0.872 +/- 0.009
Sonar	alternating *	F1	0.597 +/- 0.075	0.708 +/- 0.000	0.809 +/- 0.024
Sonar	first_half	F1	0.573 +/- 0.009	0.638 +/- 0.000	0.761 +/- 0.023
Semeion	center *	F1	0.849 +/- 0.006	0.909 +/- 0.000	0.908 +/- 0.007
Semeion	right_half	F1	0.746 +/- 0.045	0.781 +/- 0.000	0.836 +/- 0.008
Letter	alternating *	F1	0.522 +/- 0.002	0.498 +/- 0.000	0.700 +/- 0.011
Letter	first_half	F1	0.329 +/- 0.011	0.298 +/- 0.000	0.447 +/- 0.003
PenDigits	alternating_pairs *	F1	0.898 +/- 0.005	0.833 +/- 0.000	0.911 +/- 0.001
PenDigits	first_half	F1	0.819 +/- 0.015	0.703 +/- 0.000	0.827 +/- 0.008
SatImage	neighbor_to_center *	F1	0.799 +/- 0.020	0.806 +/- 0.000	0.868 +/- 0.008
SatImage	spectral_halves	F1	0.761 +/- 0.023	0.746 +/- 0.000	0.836 +/- 0.003
Human activity (HAR)	time_to_frequency *	F1	0.796 +/- 0.014	0.929 +/- 0.000	0.916 +/- 0.001
Human activity (HAR)	alternating	F1	0.773 +/- 0.015	0.897 +/- 0.000	0.915 +/- 0.014
Dry Bean	alternating *	F1	0.914 +/- 0.002	0.914 +/- 0.000	0.916 +/- 0.003
Dry Bean	first_half	F1	0.902 +/- 0.007	0.920 +/- 0.000	0.899 +/- 0.006
ISOLET	alternating *	F1	0.748 +/- 0.005	0.929 +/- 0.000	0.932 +/- 0.006
ISOLET	first_half	F1	0.537 +/- 0.030	0.668 +/- 0.000	0.642 +/- 0.012
Lorenz trajectories	past24_future8 *	RMSE	0.850 +/- 0.072	2.384 +/- 0.000	0.589 +/- 0.002
Lorenz trajectories	past16_future16	RMSE	1.297 +/- 0.057	4.934 +/- 0.000	0.782 +/- 0.010
Mackey-Glass delay	past24_future8 *	RMSE	0.089 +/- 0.005	0.078 +/- 0.000	0.087 +/- 0.001
Mackey-Glass delay	past16_future16	RMSE	0.131 +/- 0.003	0.141 +/- 0.000	0.126 +/- 0.001
Switching VAR	past24_future8 *	RMSE	0.183 +/- 0.001	0.185 +/- 0.000	0.179 +/- 0.000
Switching VAR	past16_future16	RMSE	0.188 +/- 0.001	0.190 +/- 0.000	0.186 +/- 0.000
Chirp + seasonal	past24_future8 *	RMSE	0.211 +/- 0.015	0.162 +/- 0.000	0.197 +/- 0.000

Dataset	Policy (* primary)	Metric	JEPA	Raw linear	Extra Trees
Chirp + seasonal	past16_future16	RMSE	0.279 +/- 0.010	0.283 +/- 0.000	0.265 +/- 0.001
Coupled oscillators	past24_future8 *	RMSE	0.138 +/- 0.007	0.036 +/- 0.000	0.131 +/- 0.001
Coupled oscillators	past16_future16	RMSE	0.147 +/- 0.019	0.064 +/- 0.000	0.145 +/- 0.001
Nonlinear AR	past24_future8 *	RMSE	0.115 +/- 0.008	0.098 +/- 0.000	0.121 +/- 0.000
Nonlinear AR	past16_future16	RMSE	0.115 +/- 0.008	0.103 +/- 0.000	0.124 +/- 0.000
Nonlinear multiview	view_a_to_view_b *	F1	0.880 +/- 0.011	0.849 +/- 0.000	0.919 +/- 0.007
Nonlinear multiview	alternating	F1	0.880 +/- 0.014	0.846 +/- 0.000	0.901 +/- 0.008
Hierarchical classes	view_a_to_view_b *	F1	0.963 +/- 0.005	0.950 +/- 0.000	0.961 +/- 0.003
Hierarchical classes	alternating	F1	0.962 +/- 0.004	0.960 +/- 0.000	0.964 +/- 0.002
Sparse interactions	view_a_to_view_b *	F1	0.578 +/- 0.018	0.665 +/- 0.000	0.657 +/- 0.011
Sparse interactions	alternating	F1	0.757 +/- 0.020	0.935 +/- 0.000	0.889 +/- 0.018
Manifold + nuisance	view_a_to_view_b *	F1	0.765 +/- 0.028	0.736 +/- 0.000	0.958 +/- 0.003
Manifold + nuisance	alternating	F1	0.843 +/- 0.031	0.908 +/- 0.000	0.953 +/- 0.009

Original snapshot and provenance

Dataset	Original primary policy	JEPA	Raw linear	Extra Trees
Wine	alternating	0.914 +/- 0.000	0.941 +/- 0.000	0.980 +/- 0.018
Digits	center	0.886 +/- 0.008	0.875 +/- 0.000	0.913 +/- 0.008
Original sensors	past16_future8	0.321 +/- 0.021	0.259 +/- 0.000	0.358 +/- 0.004

Wine and Digits use macro F1; original sensors use RMSE. The 18 original runs are preserved with their historical runtime snapshot in results/original_runtime.zip. No current-source-equivalence claim is made for those older measurements.

Public sources and attribution

Breast cancer (WDBC): Wolberg, W., Mangasarian, O., Street, N. & Street, W. (1993). Breast Cancer Wisconsin (Diagnostic). <https://doi.org/10.24432/C5DW2B>
<https://archive.ics.uci.edu/dataset/17/breast+cancer+wisconsin+diagnostic>

Ionosphere: Sigillito, V., Wing, S., Hutton, L. & Baker, K. (1989). Ionosphere. <https://doi.org/10.24432/C5W01B>
<https://archive.ics.uci.edu/dataset/52/ionosphere>

Sonar: Sejnowski, T. & Gorman, R. (1988). Connectionist Bench (Sonar, Mines vs. Rocks). <https://doi.org/10.24432/C5T01Q>
<https://archive.ics.uci.edu/dataset/151/connectionist+bench+sonar+mines+vs+rocks>

Semeion: Semeion Handwritten Digit (1998). UCI Machine Learning Repository. <https://doi.org/10.24432/C5SC8V>
<https://archive.ics.uci.edu/dataset/178/semeion+handwritten+digit>

Letter: Slate, D. (1991). Letter Recognition. <https://doi.org/10.24432/C5ZP40>
<https://archive.ics.uci.edu/dataset/59/letter+recognition>

PenDigits: Alpaydin, E. & Alimoglu, F. (1996). Pen-Based Recognition of Handwritten Digits. <https://doi.org/10.24432/C5MG6K>
<https://archive.ics.uci.edu/dataset/81/pen+based+recognition+of+handwritten+digits>

SatImage: Srinivasan, A. (1993). Statlog (Landsat Satellite). <https://doi.org/10.24432/C55887>
<https://archive.ics.uci.edu/dataset/146/statlog+landsat+satellite>

Human activity (HAR): Reyes-Ortiz, J., Anguita, D., Ghio, A., Oneto, L. & Parra, X. (2013). Human Activity Recognition Using Smartphones. <https://doi.org/10.24432/C54S4K>
<https://archive.ics.uci.edu/dataset/240/human+activity+recognition+using+smartphones>

Dry Bean: Dry Bean (2020). UCI Machine Learning Repository. Koklu, M. & Ozkan, I. A. (2020), Computers and Electronics in Agriculture 174, 105507. <https://doi.org/10.24432/C50S4B>
<https://archive.ics.uci.edu/dataset/602/dry+bean+dataset>

ISOLET: Cole, R. & Fandy, M. (1991). ISOLET. <https://doi.org/10.24432/C51G69>
<https://archive.ics.uci.edu/dataset/54/isolet>

Synthetic-source boundary

Synthetic datasets are generated locally with recorded equations, seeds, and parameters. Named dynamics such as Lorenz or Mackey-Glass identify the implemented mechanism; they do not establish calibrated realism or exact reproduction of another benchmark. Mechanism reference links and full parameter records are in EXPANDED_REPORT.md.

Evidence files

results/expanded_validation.json; results/expanded_summary.json; artifacts/expanded/results.json; results/expanded_test_report.xml; EXPANDED_REPORT.md.

Expanded result SHA-256: 8f223e56d42f1a23bda989cdfa1419c2e261843627fd1e4db8f007dc1c8d90a5